MATH 350-2 Advanced Calculus

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Outline

1. Real Analysis Lecture 15
   - Limit of a function
   - Continuity

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   - Limit of a function
   - Continuity

Back in calculus

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as x gets closer and closer to a,

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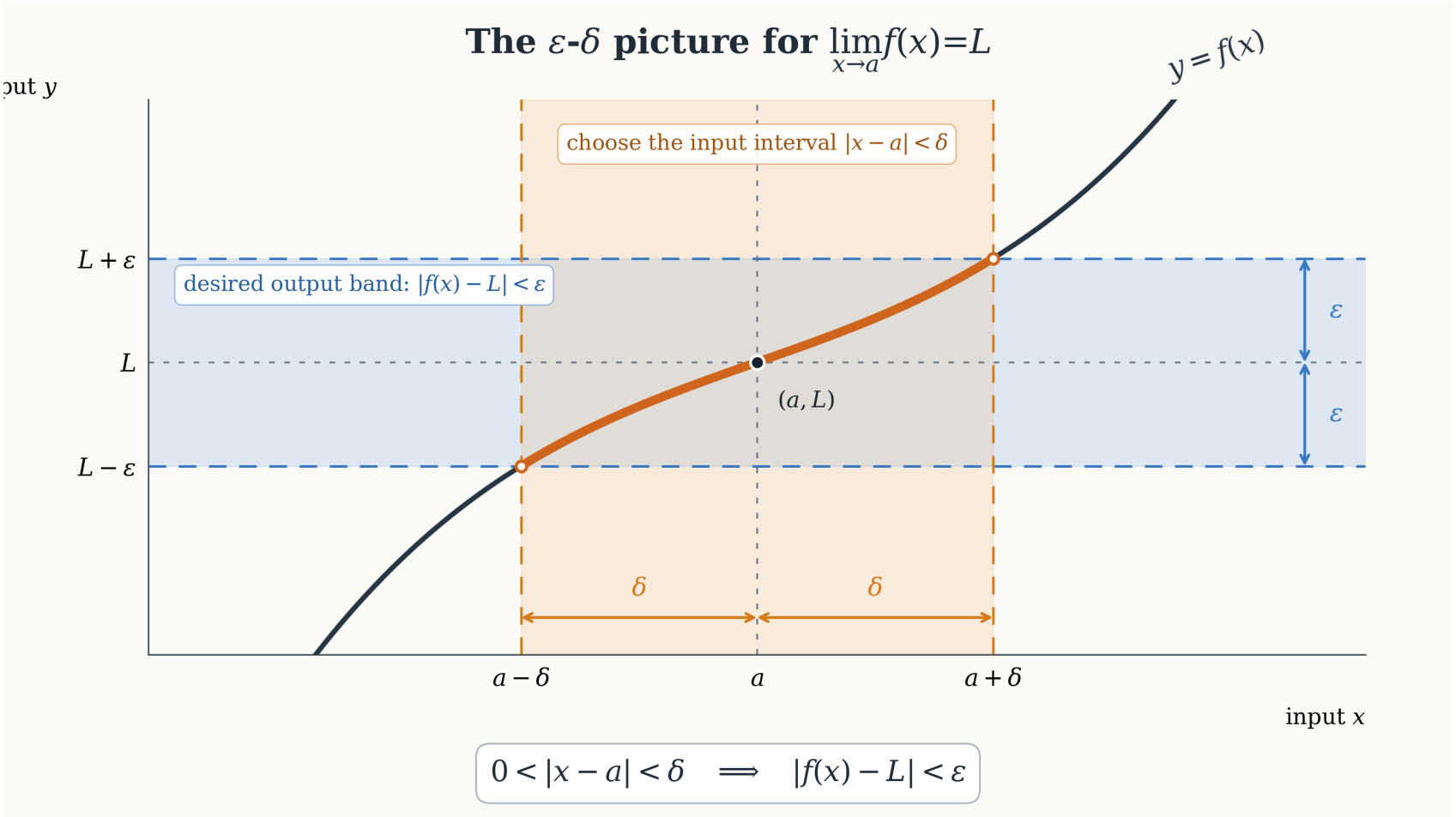
Back in calculus

The limit as $x \to a$ of $f(x)$ is L means:
as x gets closer and closer to a, $f(x)$ gets closer and closer to L.
Rigorous: For all $\epsilon > 0$ there exists $\delta > 0$ such that

$$0 < |x - a| < \delta \quad \Rightarrow \quad |f(x) - L| < \epsilon.$$

Graphical intuition



Challenge

Problem

Prove that

$$\lim_{x \to 5} 2x - 7 = 3$$

Solution

Proof.

Solution

Proof.

Let $\epsilon > 0$.

Solution

Proof.

Let $\epsilon > 0$.

Choose $\delta = \epsilon/2$.

Solution

Proof.

Let $\epsilon > 0$.

Choose $\delta = \epsilon/2$. Then for $0 < |x - 5| < \delta$ we have

$$|2x - 7 - 3| = |2x - 10| = 2|x - 5| = 2\epsilon/2 = \epsilon$$

Solution

Proof.

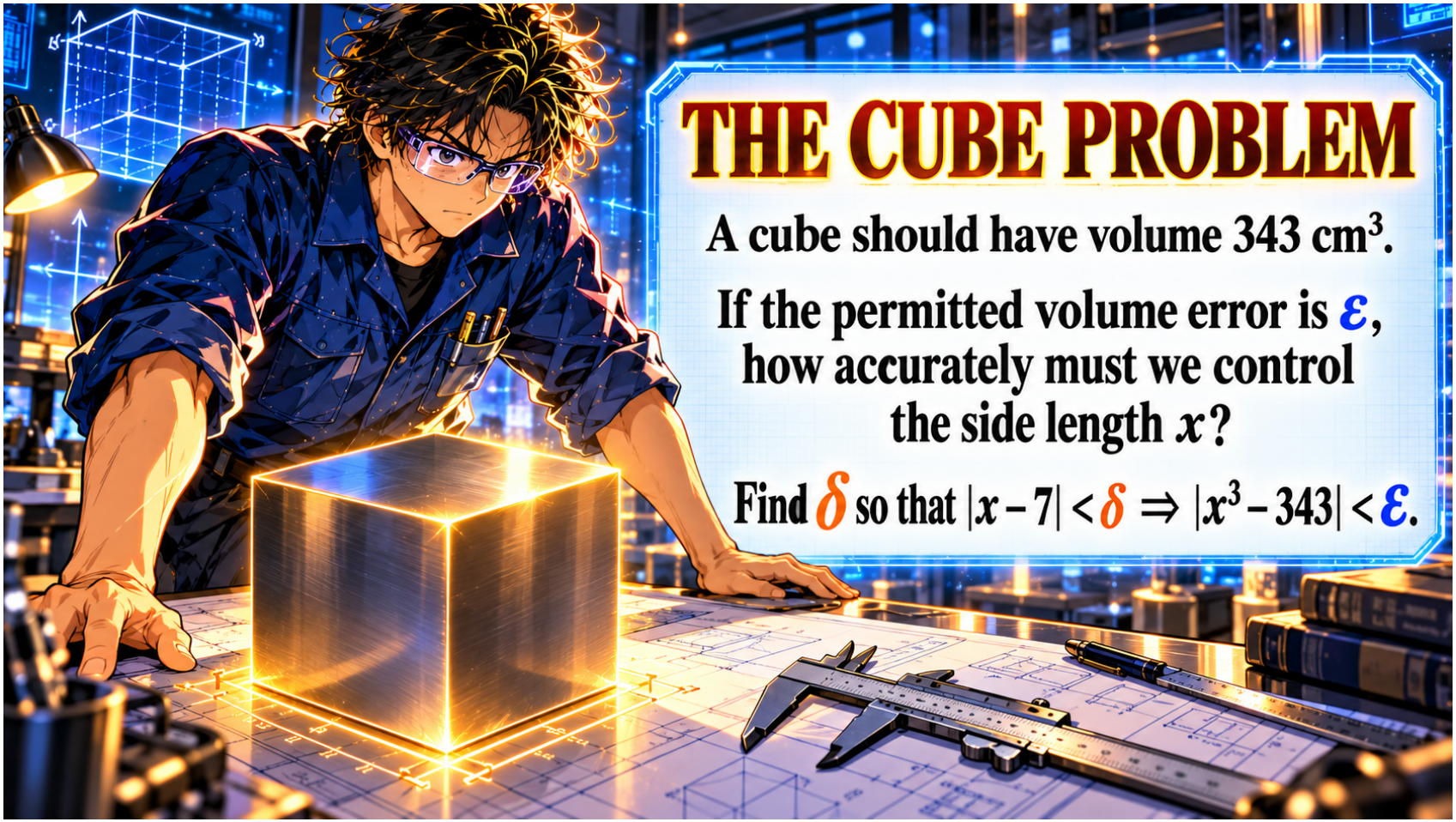
Let $\epsilon > 0$.

Choose $\delta = \epsilon/2$. Then for $0 < |x - 5| < \delta$ we have

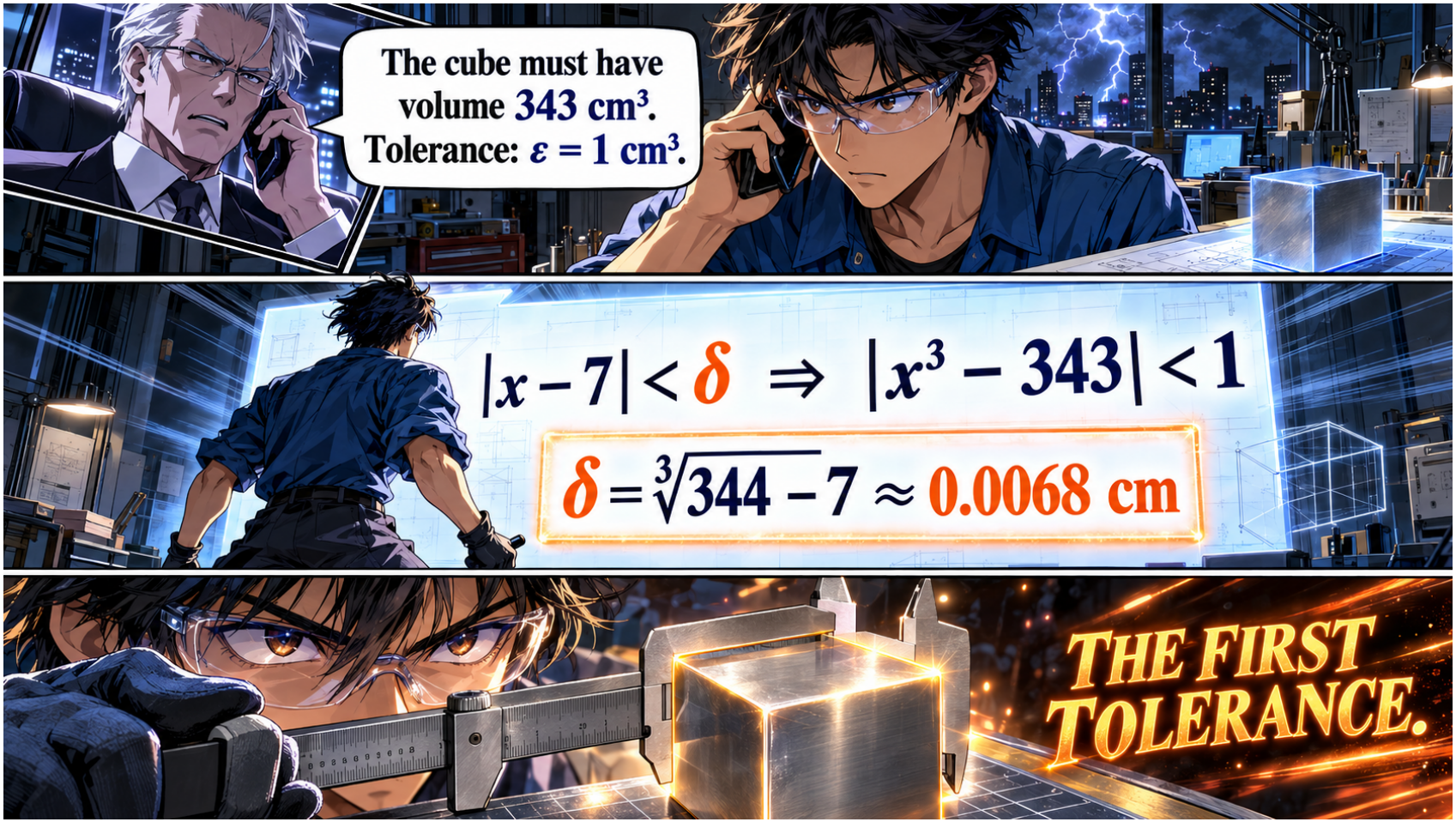
$$|2x - 7 - 3| = |2x - 10| = 2|x - 5| = 2\epsilon/2 = \epsilon$$

Since $\epsilon > 0$ was arbitrary, this proves $\lim_{x \to 5} 2x - 7 = 3$. □

Engineering interpretation



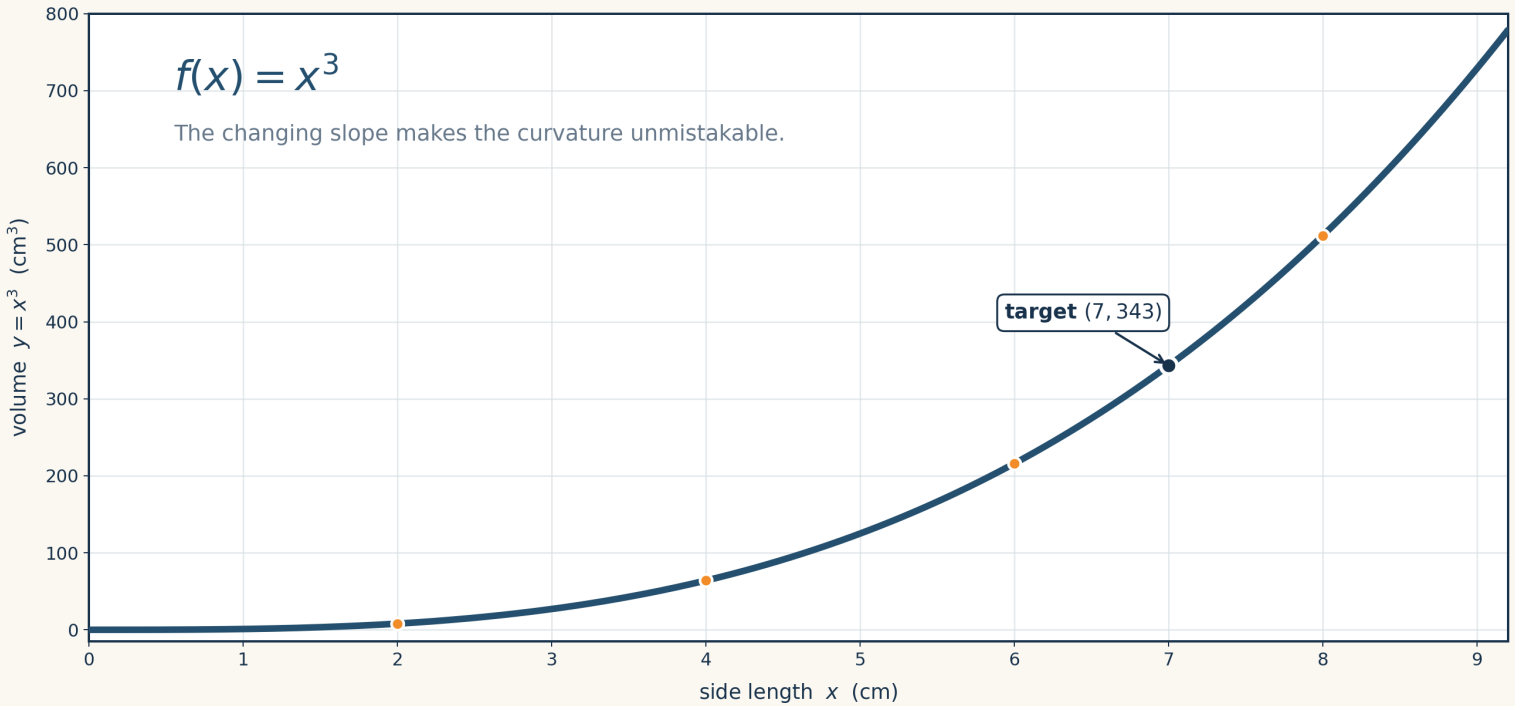
Engineering interpretation



Engineering interpretation

1. Start with the whole cubic curve

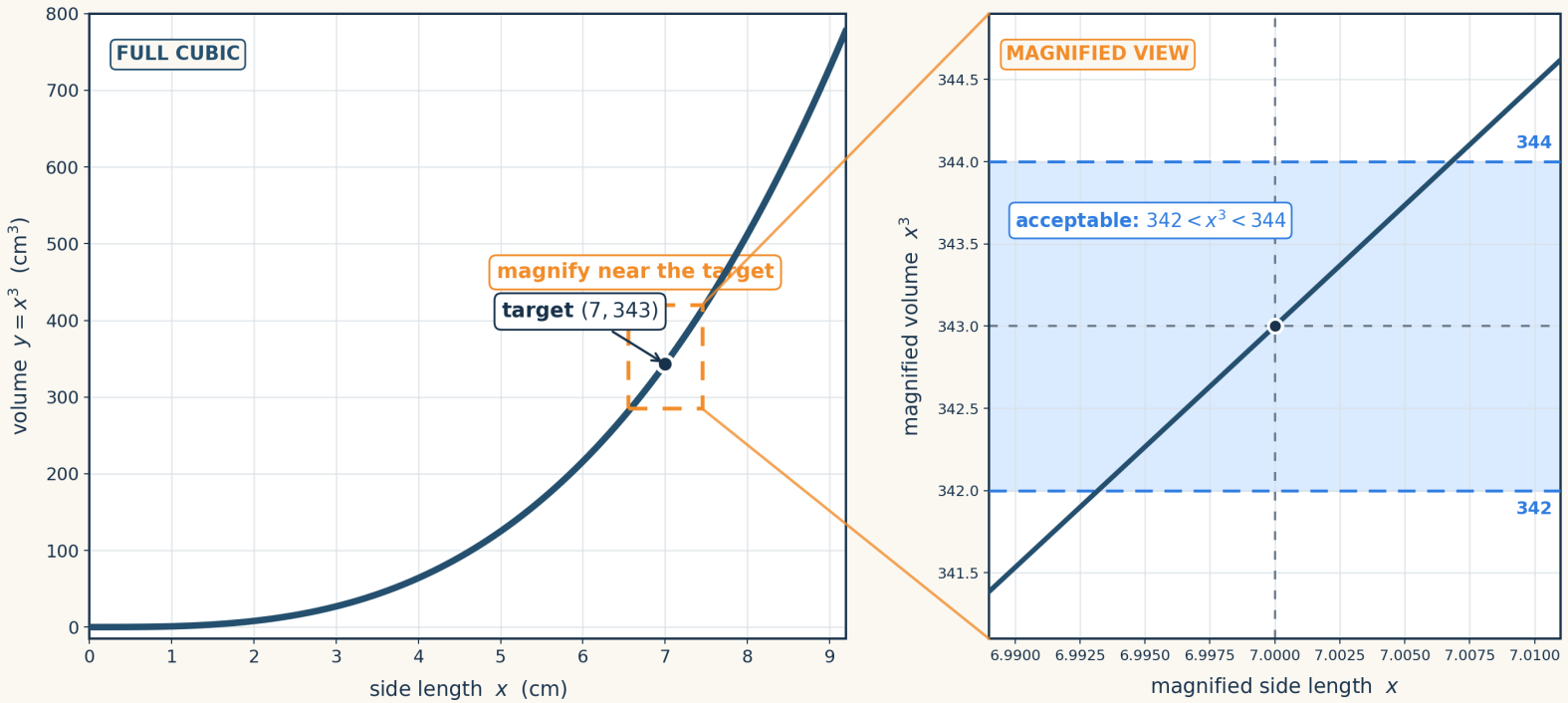
Keep a broad view so the nonlinear shape of f(x)=x³ remains visible.



Engineering interpretation

2. Draw the permitted output error

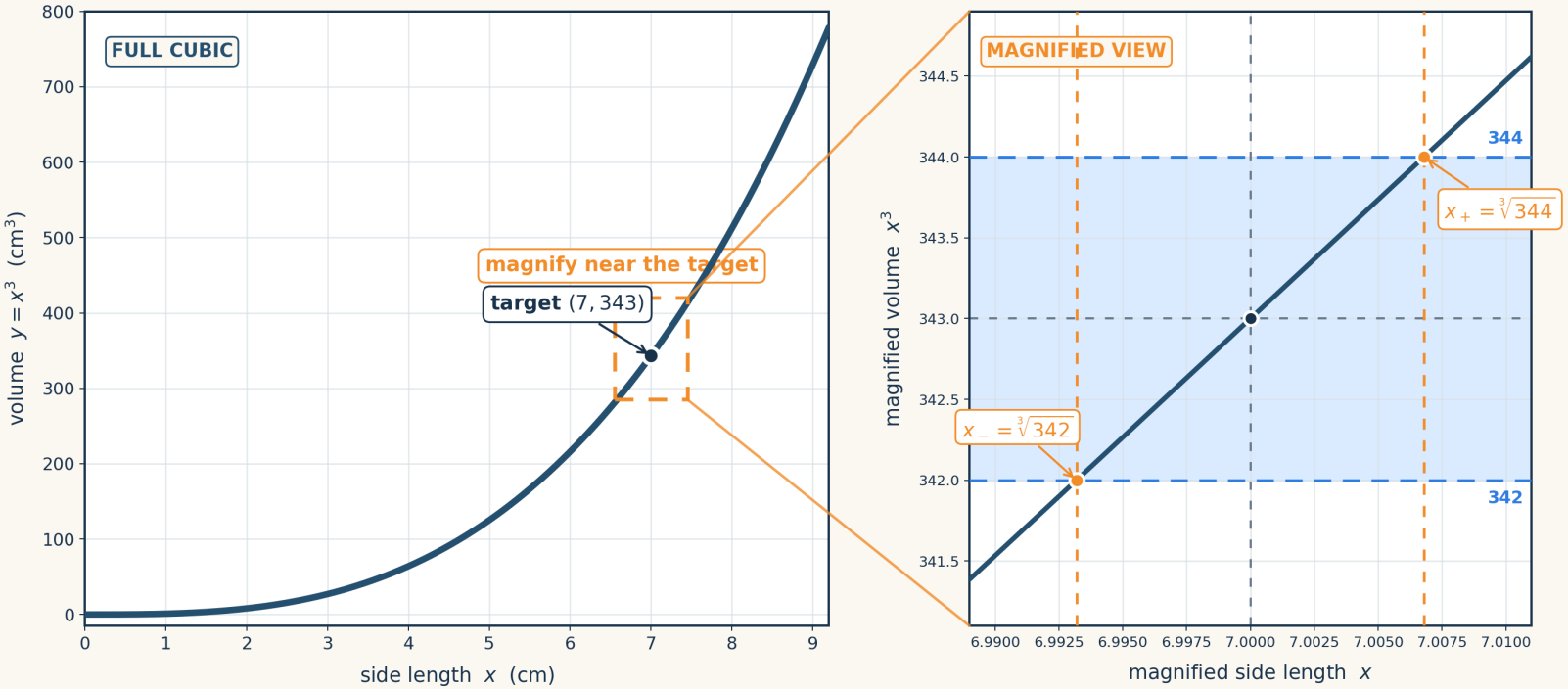
The full graph supplies the geometry; the inset makes the narrow ε=1 band readable.



Engineering interpretation

3. Find where the cubic enters and leaves the ε-band

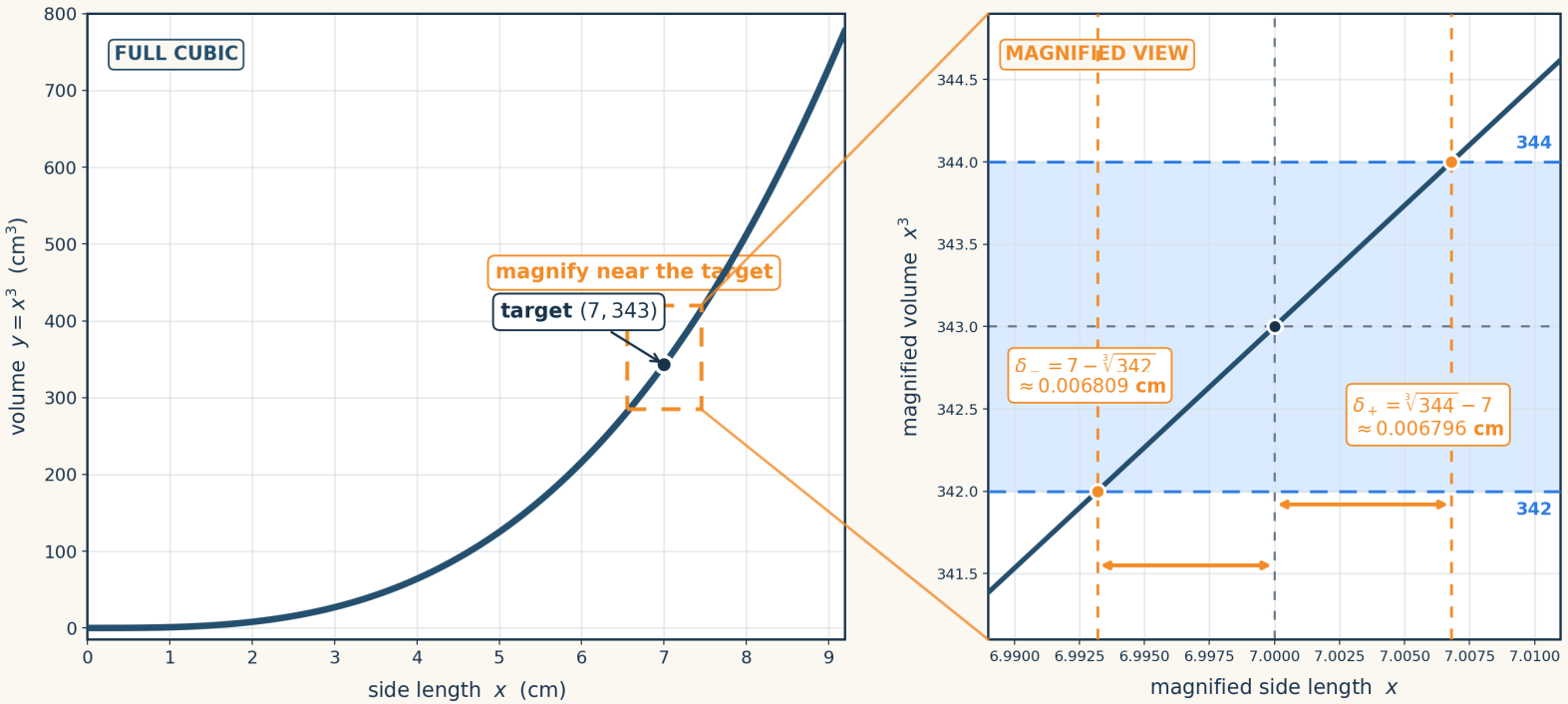
The horizontal boundaries meet $y=x^3$ at $x_-=\sqrt[3]{342}$ and $x_+=\sqrt[3]{344}$.



Engineering interpretation

4. Measure both horizontal distances

The available distance is slightly different on the two sides because the curve bends.



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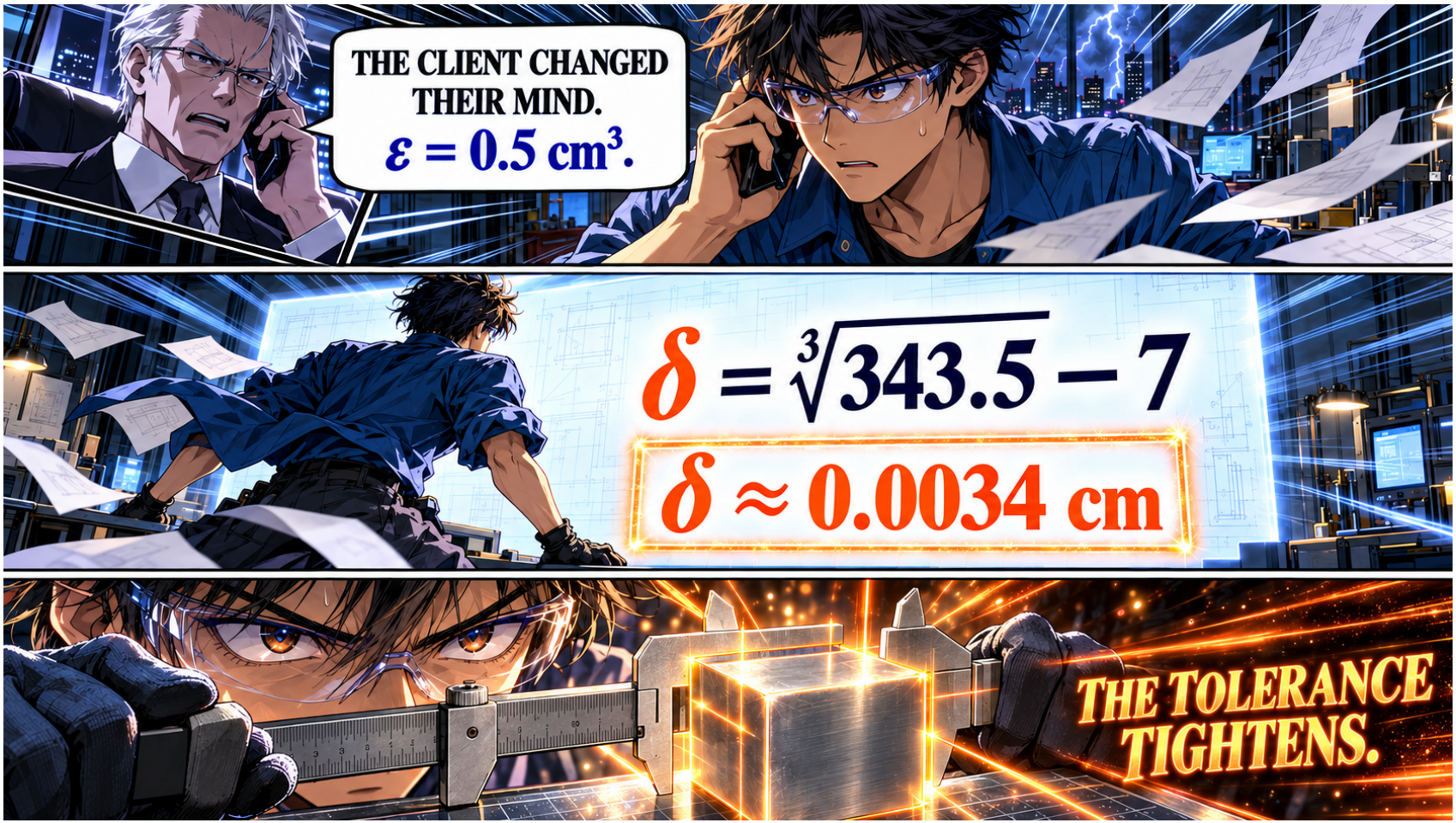

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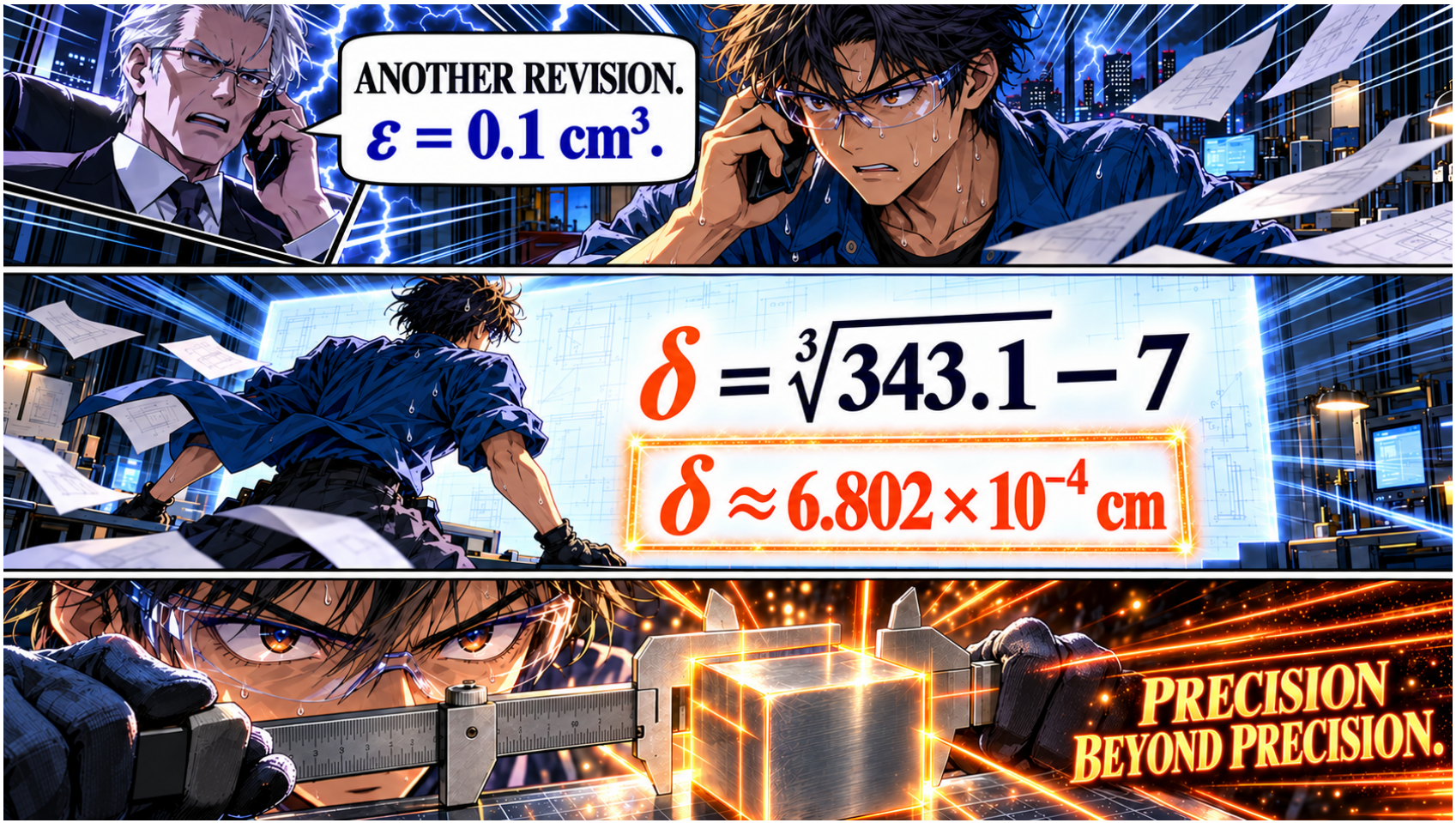

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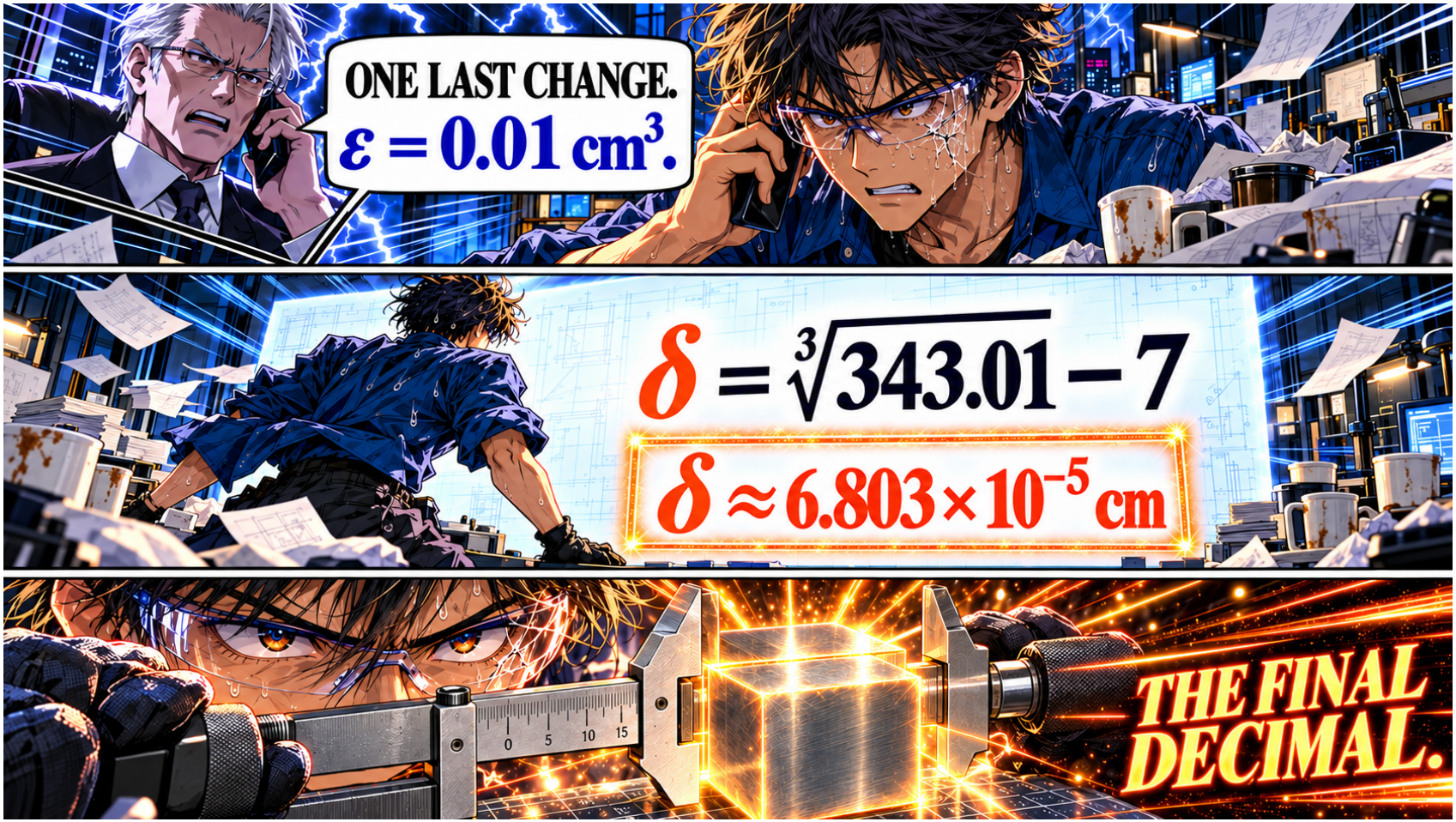

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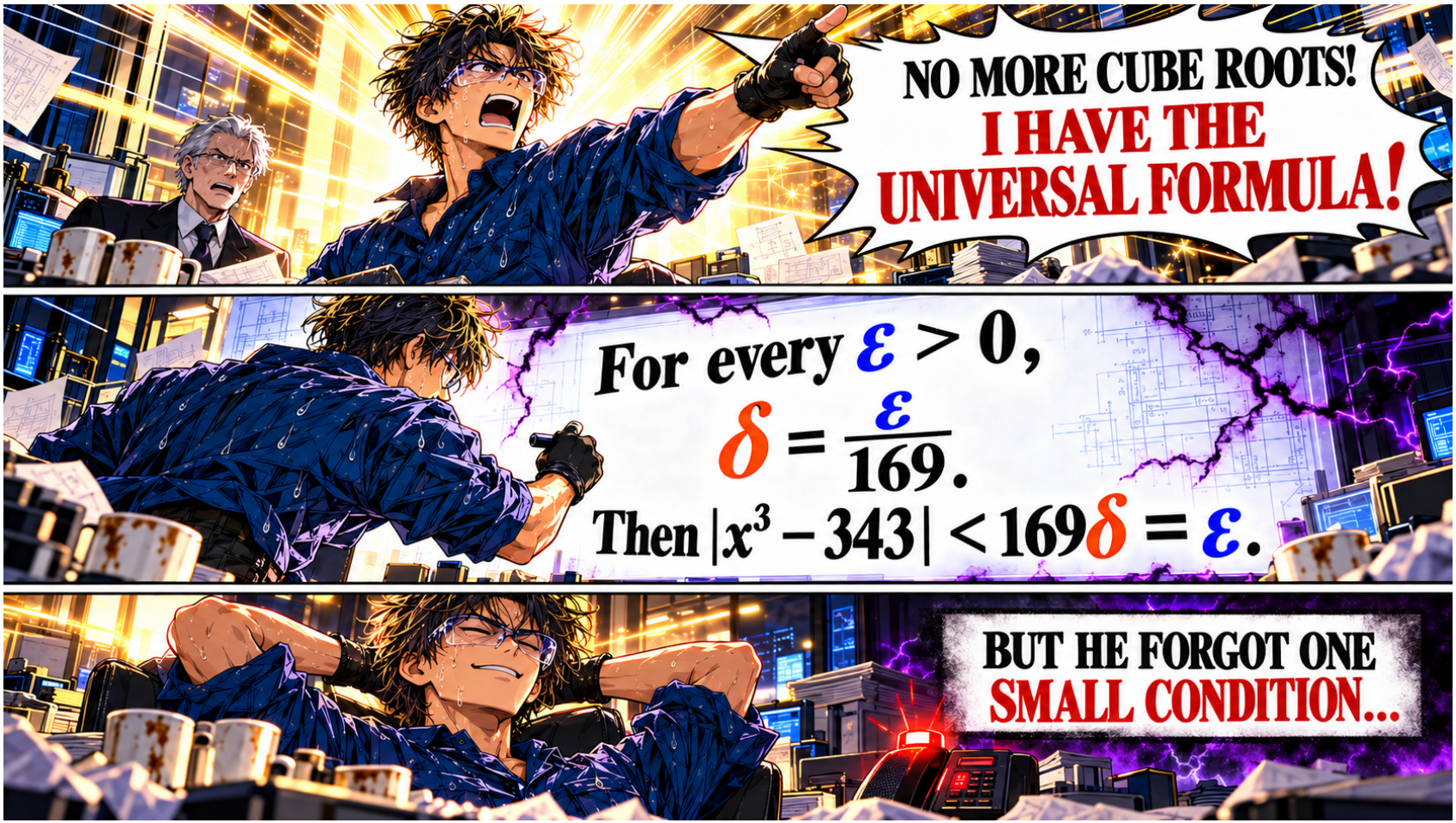

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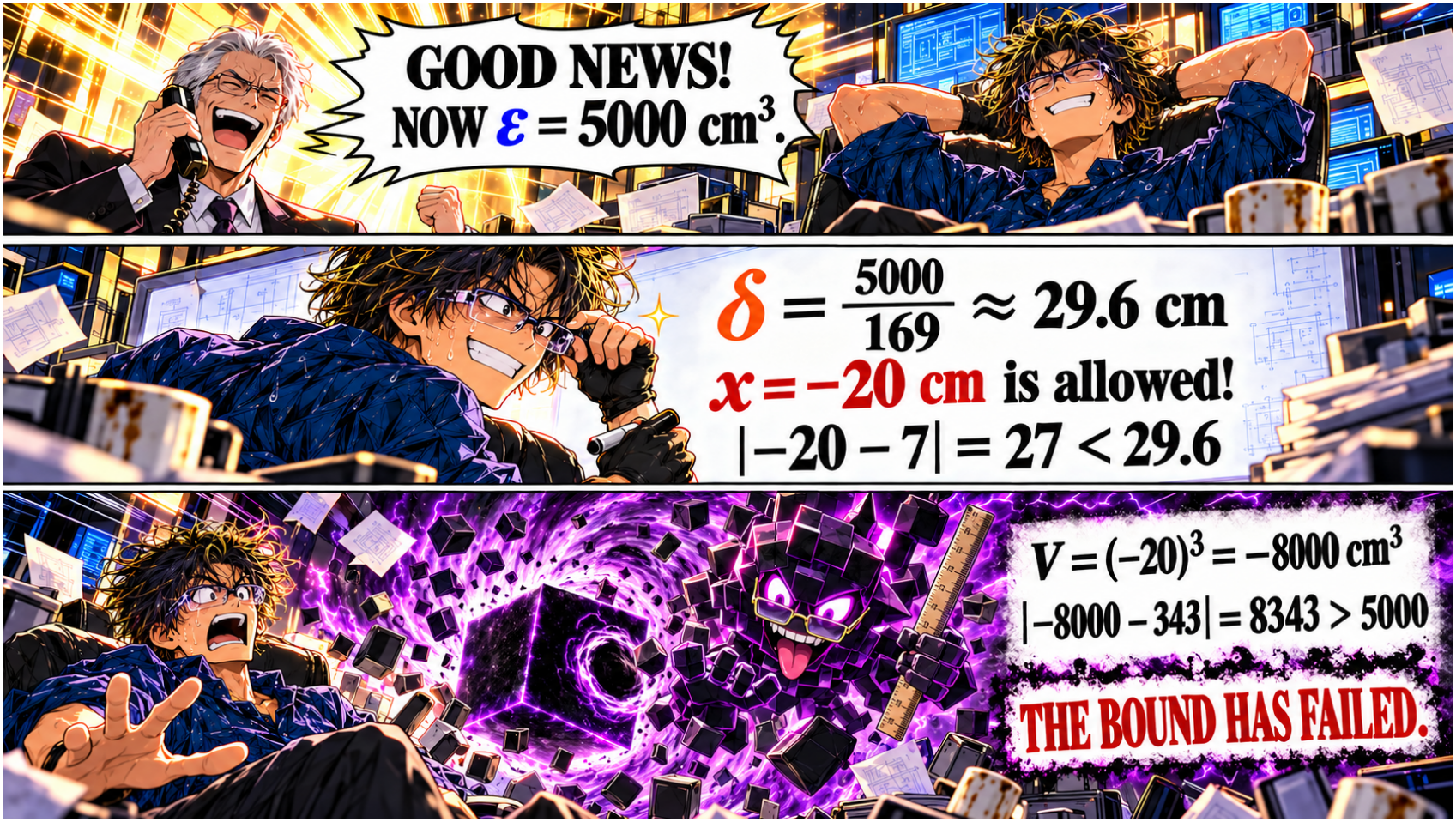

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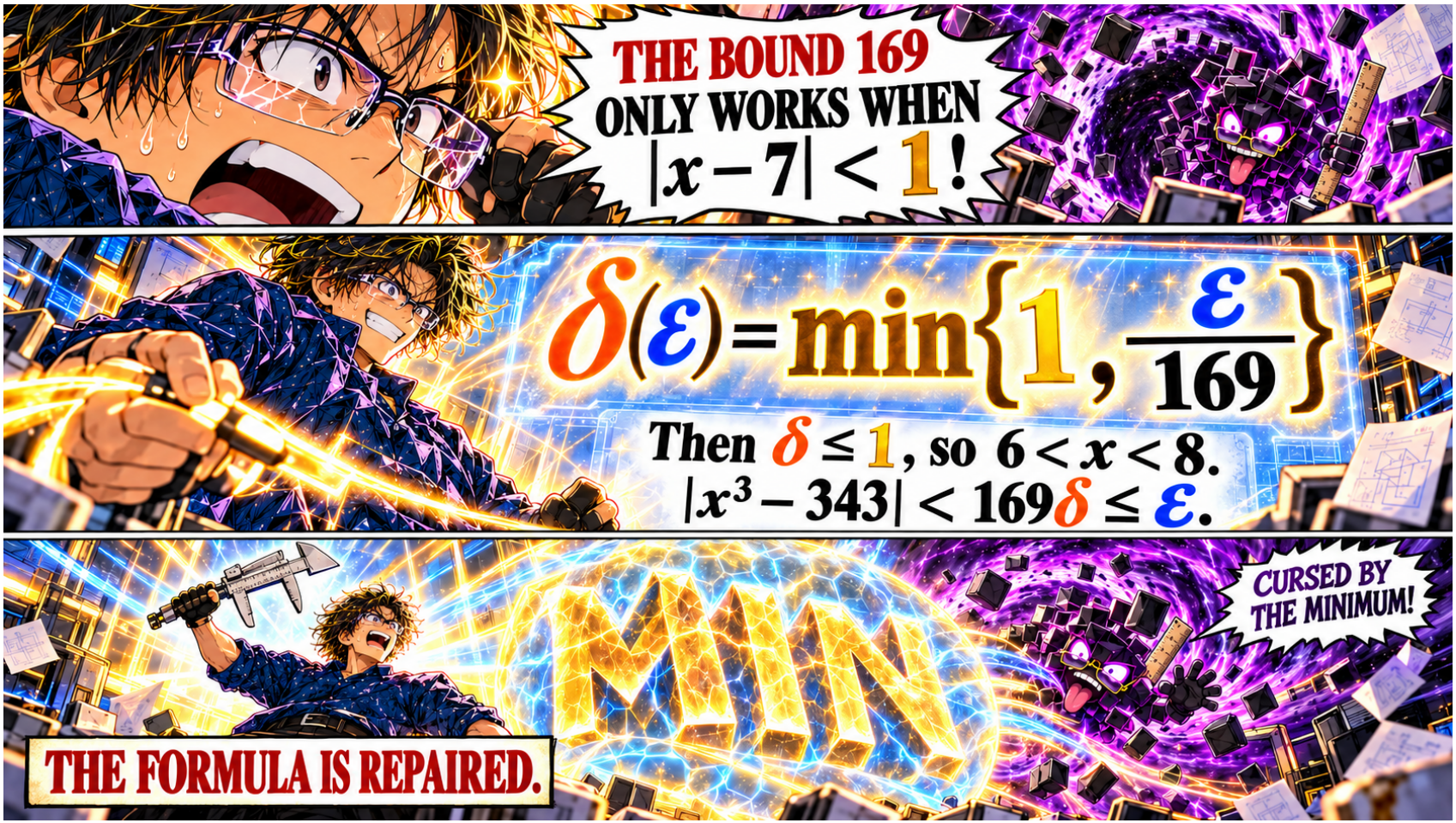

Formal Example

Problem

Prove that

$$\lim_{x \to 7} x^3 = 343$$

Solution

Proof.

Solution

Proof.

Let $\epsilon > 0$.

Solution

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Choose $\delta = \min\left(1, \frac{\epsilon}{169}\right)$.

Solution

Proof.

Let $\epsilon > 0$.

Choose $\delta = \min\left(1, \frac{\epsilon}{169}\right)$. Then for $0 < |x - 7| < \delta$ we have $|x^2 + 7x + 49| \leq 8^2 + 7 \cdot 8 + 49 = 169$.

Solution

Proof.

Let $\epsilon > 0$.
Choose $\delta = \min\left(1, \frac{\epsilon}{169}\right)$. Then for $0 < |x-7| < \delta$ we have $|x^2 + 7x + 49| \leq 8^2 + 7 \cdot 8 + 49 = 169$.
Therefore

$$\left|x^3 - 343\right| = |(x-7)(x^2+7x+49)| \leq 169|x-7| \leq \epsilon$$

Solution

Proof.

Let $\epsilon > 0$.
Choose $\delta = \min\left(1, \frac{\epsilon}{169}\right)$. Then for $0 < |x-7| < \delta$ we have $|x^2 + 7x + 49| \leq 8^2 + 7 \cdot 8 + 49 = 169$.
Therefore

$$\left|x^3 - 343\right| = |(x-7)(x^2+7x+49)| \leq 169|x-7| \leq \epsilon$$

Since $\epsilon > 0$ was arbitrary, this proves $\lim_{x \to 7} x^3 = 343$. □

Challenge

Problem

Prove that

$$\lim_{x \to 3} \frac{1}{x^2} = \frac{1}{9}$$

Solution

Proof.

Solution

Proof.

Let $\epsilon > 0$.

Solution

Proof.

Let $\epsilon > 0$.
Choose $\delta = \min\left(1, \frac{36\epsilon}{7}\right)$.

Solution

Proof.

Let $\epsilon > 0$.
Choose $\delta = \min\left(1, \frac{36\epsilon}{7}\right)$. Then for $0 < |x-3| < \delta$ we have $|x+3| < 4$ and $x^2 > 4$.

Solution

Proof.

Let $\epsilon > 0$.
Choose $\delta = \min\left(1, \frac{36\epsilon}{7}\right)$. Then for $0 < |x-3| < \delta$ we have $|x+3| < 4$ and $x^2 > 4$.
Therefore

$$\left|\frac{1}{x^2} - \frac{1}{9}\right| = \frac{|x^2-9|}{9x^2}$$
$$= \frac{|x+3|}{9x^2}|x-3| \leq \frac{7}{9\cdot 4}|x-3| < \epsilon.$$

Solution

Proof.

Let $\epsilon > 0$.
Choose $\delta = \min\left(1, \frac{36\epsilon}{7}\right)$. Then for $0 < |x-3| < \delta$ we have $|x+3| < 4$ and $x^2 > 4$.
Therefore

$$\left|\frac{1}{x^2} - \frac{1}{9}\right| = \frac{|x^2-9|}{9x^2}$$
$$= \frac{|x+3|}{9x^2}|x-3| \leq \frac{7}{9\cdot 4}|x-3| < \epsilon.$$

Since $\epsilon > 0$ was arbitrary, this proves $\lim_{x\to 3} \frac{1}{x^2} = \frac{1}{9}$. □

Example

Prove that

$$\lim_{x \to 4} \frac{\sqrt{x} - 2}{x - 4} = \frac{1}{4}$$

Limit of a function

Let (S, d_S) and (T, d_T) be metric spaces and $A \subseteq S$.

Limit of a function

Let (S, d_S) and (T, d_T) be metric spaces and $A \subseteq S$. Also, let $f : A \to T$ be a function.

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Definition

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Definition

Then for any accumulation point p of A and element $L \in T$, we say that the **limit as** x **approaches** p **of** $f(x)$ **is** L

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$$\lim_{x \to p} f(x) = L$$

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Let (S, d_S) and (T, d_T) be metric spaces and $A \subseteq S$. Also, let $f : A \to T$ be a function.

Definition

Then for any accumulation point p of A and element $L \in T$, we say that the **limit as** x **approaches** p **of** $f(x)$ **is** L and write

$$\lim_{x \to p} f(x) = L$$

if for all $\epsilon > 0$ there exists $\delta > 0$ such that

$$0 < d_S(x, p) < \delta \;\Rightarrow\; d_T(f(x), L) < \epsilon.$$

Limit of a function

Theorem

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Let (S, d_S) and (T, d_T) be metric spaces, $A \subseteq S$, and $f : A \to T$ be a function.

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Limit of a function

Theorem

Let (S, d_S) and (T, d_T) be metric spaces, $A \subseteq S$, and $f : A \to T$ be a function. Then

$$\lim_{x \to a} f(x) = L$$

if and only if

$$\lim_{n \to \infty} f(x_n) = L$$

for all sequences $\{x_n\}$ of values in $A \backslash \{a\}$ which converge to a.

Challenge!

Consider the function

$$f : \mathbb{R}^2 \to \mathbb{R}$$

between metric spaces with the Euclidean metric defined by

$$f(x, y) = \frac{x^2 y}{x^4 + y^2}.$$

Challenge!

Consider the function

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Question

Challenge!

Consider the function

$$f : \mathbb{R}^2 \to \mathbb{R}$$

between metric spaces with the Euclidean metric defined by

$$f(x, y) = \frac{x^2 y}{x^4 + y^2}.$$

Question

What is the limit as $(x, y) \to (0, 0)$?

Challenge!

Consider the sequence $\{(x_n, y_n)\}$ defined by

$$(x_n, y_n) = (1/n, 0).$$

Challenge!

Consider the sequence $\{(x_n, y_n)\}$ defined by

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Then

$$\lim_{n\to\infty}(x_n, y_n) = \lim_{n\to\infty}(1/n, 0) = (0, 0)$$

Challenge!

Consider the sequence $\{(x_n, y_n)\}$ defined by

$$(x_n, y_n) = (1/n, 0).$$

Then

$$\lim_{n\to\infty}(x_n, y_n) = \lim_{n\to\infty}(1/n, 0) = (0, 0)$$

and

$$\lim_{n\to\infty} f(x_n, y_n) = \lim_{n\to\infty} 0 = 0.$$

Challenge!

Consider the sequence $\{(x_n, y_n)\}$ defined by

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Then

$$\lim_{n\to\infty}(x_n, y_n) = \lim_{n\to\infty}(1/n, 1/n) = (0, 0)$$

and

$$\lim_{n\to\infty} f(x_n, y_n) = \lim_{n\to\infty} \frac{1/n^3}{1/n^4 + 1/n^2} = \lim_{n\to\infty} \frac{1/n}{1/n^2 + 1} = 0.$$

Challenge!

Consider the sequence $\{(x_n, y_n)\}$ defined by

$$(x_n, y_n) = (1/n, 1/n^2).$$

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$$(x_n, y_n) = (1/n, 1/n^2).$$

Then

$$\lim_{n\to\infty}(x_n, y_n) = \lim_{n\to\infty}(1/n, 1/n^2) = (0, 0)$$

and

$$\lim_{n\to\infty} f(x_n, y_n) = \lim_{n\to\infty} \frac{1/n^4}{1/n^4 + 1/n^4} = \lim_{n\to\infty} \frac{1}{2} = \frac{1}{2}.$$

Challenge!

Consider the sequence $\{(x_n, y_n)\}$ defined by

$$(x_n, y_n) = (1/n, 1/n^2).$$

Then

$$\lim_{n\to\infty}(x_n, y_n) = \lim_{n\to\infty}(1/n, 1/n^2) = (0, 0)$$

and

$$\lim_{n\to\infty} f(x_n, y_n) = \lim_{n\to\infty} \frac{1/n^4}{1/n^4 + 1/n^4} = \lim_{n\to\infty} \frac{1}{2} = \frac{1}{2}.$$

The limit DOES NOT EXIST!

Limit of a function

Proof.

($\Rightarrow$) Suppose that $\lim_{x \to a} f(x) = L$ and let $\{x_n\}$ be a sequence of values in $A \backslash \{a\}$ which converges to a.

Limit of a function

Proof.

($\Rightarrow$) Suppose that $\lim_{x \to a} f(x) = L$ and let $\{x_n\}$ be a sequence of values in $A \backslash \{a\}$ which converges to a.
Let $\epsilon > 0$.

Limit of a function

Proof.

($\Rightarrow$) Suppose that $\lim_{x\to a} f(x) = L$ and let $\{x_n\}$ be a sequence of values in $A\backslash\{a\}$ which converges to a.
Let $\epsilon > 0$.
Then there exists $\delta > 0$ such that $0 < d_S(x, a) < \delta$ implies $d_T(f(x), L) < \epsilon$.

Limit of a function

Proof.

($\Rightarrow$) Suppose that $\lim_{x\to a} f(x) = L$ and let $\{x_n\}$ be a sequence of values in $A\backslash\{a\}$ which converges to a.
Let $\epsilon > 0$.
Then there exists $\delta > 0$ such that $0 < d_S(x, a) < \delta$ implies $d_T(f(x), L) < \epsilon$.
Also, there exists $N \in \mathbb{Z}_+$ such that $n \geq N$ implies $d_S(x_n, a) < \delta$ for all $n \geq N$.

Limit of a function

Proof.

($\Rightarrow$) Suppose that $\lim_{x\to a} f(x) = L$ and let $\{x_n\}$ be a sequence of values in $A\backslash\{a\}$ which converges to a.
Let $\epsilon > 0$.
Then there exists $\delta > 0$ such that $0 < d_S(x, a) < \delta$ implies $d_T(f(x), L) < \epsilon$.
Also, there exists $N \in \mathbb{Z}_+$ such that $n \geq N$ implies $d_S(x_n, a) < \delta$ for all $n \geq N$.
Therefore since $x_n \neq a$ for all n, we have that

$$n \geq N \;\Rightarrow\; 0 < d_S(x_n, a) < \delta \;\Rightarrow\; d_T(f(x_n), L) < \epsilon.$$

Limit of a function

Proof.

($\Rightarrow$) Suppose that $\lim_{x\to a} f(x) = L$ and let $\{x_n\}$ be a sequence of values in $A\backslash\{a\}$ which converges to a.
Let $\epsilon > 0$.
Then there exists $\delta > 0$ such that $0 < d_S(x, a) < \delta$ implies $d_T(f(x), L) < \epsilon$.
Also, there exists $N \in \mathbb{Z}_+$ such that $n \geq N$ implies $d_S(x_n, a) < \delta$ for all $n \geq N$.
Therefore since $x_n \neq a$ for all n, we have that

$$n \geq N \;\Rightarrow\; 0 < d_S(x_n, a) < \delta \;\Rightarrow\; d_T(f(x_n), L) < \epsilon.$$

Hence for all $n \geq N$, we know $d_T(f(x_n), L) < \epsilon$.

Limit of a function

Proof.

($\Rightarrow$) Suppose that $\lim_{x\to a} f(x) = L$ and let $\{x_n\}$ be a sequence of values in $A\backslash\{a\}$ which converges to a.
Let $\epsilon > 0$.
Then there exists $\delta > 0$ such that $0 < d_S(x, a) < \delta$ implies $d_T(f(x), L) < \epsilon$.
Also, there exists $N \in \mathbb{Z}_+$ such that $n \geq N$ implies $d_S(x_n, a) < \delta$ for all $n \geq N$.
Therefore since $x_n \neq a$ for all n, we have that

$$n \geq N \;\Rightarrow\; 0 < d_S(x_n, a) < \delta \;\Rightarrow\; d_T(f(x_n), L) < \epsilon.$$

Hence for all $n \geq N$, we know $d_T(f(x_n), L) < \epsilon$.
Since $\epsilon > 0$ was arbitrary, this proves $\lim_{n\to\infty} f(x_n) = L$. □

Limit of a function

Proof.

($\Leftarrow$) Suppose that for every sequence $\{x_n\}$ of values in $A\backslash\{a\}$ which converges to a, we have $\lim_{n\to\infty} f(x_n) = L$.

Limit of a function

Proof.

($\Leftarrow$) Suppose that for every sequence $\{x_n\}$ of values in $A\backslash\{a\}$ which converges to a, we have $\lim_{n\to\infty} f(x_n) = L$. We must show that $\lim_{x\to a} f(x) = L$.

Limit of a function

Proof.

($\Leftarrow$) Suppose that for every sequence $\{x_n\}$ of values in $A\backslash\{a\}$ which converges to a, we have $\lim_{n\to\infty} f(x_n) = L$. We must show that $\lim_{x\to a} f(x) = L$.
To see this, we assume otherwise.

Limit of a function

Proof.

($\Leftarrow$) Suppose that for every sequence $\{x_n\}$ of values in $A\backslash\{a\}$ which converges to a, we have $\lim_{n\to\infty} f(x_n) = L$. We must show that $\lim_{x\to a} f(x) = L$.

To see this, we assume otherwise.

Then there exists $\epsilon > 0$ such that for all $\delta > 0$ there exists x with $0 < d_S(x, a) < \delta$ and $d_T(f(x), L) \geq \epsilon$.

Limit of a function

Proof.

($\Leftarrow$) Suppose that for every sequence $\{x_n\}$ of values in $A\backslash\{a\}$ which converges to a, we have $\lim_{n\to\infty} f(x_n) = L$. We must show that $\lim_{x\to a} f(x) = L$.

To see this, we assume otherwise.

Then there exists $\epsilon > 0$ such that for all $\delta > 0$ there exists x with $0 < d_S(x, a) < \delta$ and $d_T(f(x), L) \geq \epsilon$.

For each $n \in \mathbb{Z}_+$, take $x_n \in A$ with $0 < d_S(x_n, a) < 1/n$ and $d_T(f(x_)n, L) \geq \epsilon$.

Limit of a function

Proof.

($\Leftarrow$) Suppose that for every sequence $\{x_n\}$ of values in $A\backslash\{a\}$ which converges to a, we have $\lim_{n\to\infty} f(x_n) = L$. We must show that $\lim_{x\to a} f(x) = L$.

To see this, we assume otherwise.

Then there exists $\epsilon > 0$ such that for all $\delta > 0$ there exists x with $0 < d_S(x, a) < \delta$ and $d_T(f(x), L) \geq \epsilon$.

For each $n \in \mathbb{Z}_+$, take $x_n \in A$ with $0 < d_S(x_n, a) < 1/n$ and $d_T(f(x_)n, L) \geq \epsilon$.

In this case, $\{x_n\}$ is a sequence of values in $A\backslash\{a\}$ which converges to a.

Limit of a function

Proof.

($\Leftarrow$) Suppose that for every sequence $\{x_n\}$ of values in $A\backslash\{a\}$ which converges to a, we have $\lim_{n\to\infty} f(x_n) = L$. We must show that $\lim_{x\to a} f(x) = L$.

To see this, we assume otherwise.

Then there exists $\epsilon > 0$ such that for all $\delta > 0$ there exists x with $0 < d_S(x, a) < \delta$ and $d_T(f(x), L) \geq \epsilon$.

For each $n \in \mathbb{Z}_+$, take $x_n \in A$ with $0 < d_S(x_n, a) < 1/n$ and $d_T(f(x_)n, L) \geq \epsilon$.

In this case, $\{x_n\}$ is a sequence of values in $A\backslash\{a\}$ which converges to a.

However, $\{f(x_n)\}$ never gets closer than ϵ to L, so it cannot converge to L.

Limit of a function

Proof.

($\Leftarrow$) Suppose that for every sequence $\{x_n\}$ of values in $A\backslash\{a\}$ which converges to a, we have $\lim_{n\to\infty} f(x_n) = L$. We must show that $\lim_{x\to a} f(x) = L$.

To see this, we assume otherwise.

Then there exists $\epsilon > 0$ such that for all $\delta > 0$ there exists x with $0 < d_S(x, a) < \delta$ and $d_T(f(x), L) \geq \epsilon$.

For each $n \in \mathbb{Z}_+$, take $x_n \in A$ with $0 < d_S(x_n, a) < 1/n$ and $d_T(f(x_)n, L) \geq \epsilon$.

In this case, $\{x_n\}$ is a sequence of values in $A\backslash\{a\}$ which converges to a.

However, $\{f(x_n)\}$ never gets closer than ϵ to L, so it cannot converge to L.

This is a contradiction. $\square$

Outline

1. Real Analysis Lecture 15
   - Limit of a function
   - Continuity

Continuity

Let (S, d_S) and (T, d_T) be metric spaces and $f : S \to T$ a function.

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Equivalently, $f(B_S(p; \delta)) \subseteq B_T(f(p); \epsilon)$. We say f is continuous on $A \subseteq S$ if $f(x)$ is continuous at every point $p \in A$. If f is continuous on S, we simply say f is continuous.

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Intuitively, continuity means

$$\lim_{n\to\infty} f(x_n) = f\left(\lim_{n\to\infty} x_n\right).$$

Compositions

Theorem

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Theorem

Suppose (S, d_S), (T, d_T), and (U, d_U) are metric spaces and $f : S \to T$ and $g : T \to U$ are functions.
If f is continuous at p and g is continuous at $f(p)$, then the composition $h = g \circ f : S \to U$ defined by $h(x) = g(f(x))$ is continuous at p.

Challenge!

Problem

Prove the previous theorem!

Preimages

Definition

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Definition

Suppose $f : S \to T$ is a function and that $Y \subseteq T$.
The **inverse image** or **preimage** of Y is the set

$$f^{-1}(Y) = \{x : x \in S, \text{ and } f(x) \in Y\}.$$

Challenge!

Problem

Suppose that $S \to T$ and $Y \subseteq Z \subseteq T$.
Prove that

$$f^{-1}(Y) \subseteq f^{-1}(Z).$$

Open sets

Theorem

Suppose (S, d_S) and (T, d_T) are metric spaces and $f : S \to T$ is a function.

Open sets

Theorem

Suppose (S, d_S) and (T, d_T) are metric spaces and $f : S \to T$ is a function.
Then f is continuous if and only if $f^{-1}(V)$ is open for all open subsets $V \subseteq T$.

Closed sets

Theorem

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Closed sets

Theorem

Suppose (S, d_S) and (T, d_T) are metric spaces and $f : S \to T$ is a function.
Then f is continuous if and only if $f^{-1}(A)$ is closed for all closed subsets $A \subseteq T$.